P100 Unit Diagnostic Report

Teensy Board (Front Display) Not Detected - Physical Impact Reported

Node (planename)	N068AV
Airline	AVA (Avianca)
Serial Number	W20041874322
Reported Cause	Physical impact to the unit
Symptom	Front display (Teensy Board) does not power on / show anything
Diagnostic Date	August 20, 2026
Diagnosis	Hardware failure - Teensy Board not detected on USB bus
Rest of Unit	Fully operational (OS boots, network/internet working normally)

Diagnostic Summary

Unit N068AV suffered a reported physical impact, after which the front display (referred to internally as the Teensy Board, which also controls the front-panel buttons and status LEDs) stopped powering on. The rest of the unit is fully functional: the OS boots normally, network connectivity and internet access work without issue, and no other subsystem shows signs of failure. Diagnostics confirm that the Teensy Board is not detected at all on the internal USB bus - not merely failing to display content, but entirely absent from the system's USB enumeration. Both the udev rules and the application-level configuration for the Teensy are correctly installed and well-formed, ruling out software or configuration as the cause. This is consistent with a physical/electrical fault in the Teensy Board or its internal USB connection resulting from the reported impact.

Diagnostic Steps Performed (in order)

1. Confirmed the rest of the unit is healthy: OS boots normally and the machine connects to the internet without issue - isolating the problem to the Teensy subsystem specifically.

2. Checked for the Teensy's expected serial device (/dev/ttyACM0): device does not exist.

3. Attempted to communicate with the Teensy via its Python control library (teensycmds.TeensyBoard 'ping' command): failed with a FileNotFoundError, since the underlying serial device is absent.

4. Checked kernel USB messages (dmesg) and the full USB device list (lsusb) for any trace of the Teensy hardware (USB vendor ID 16c0, used by PJRC/Teensy boards): no Teensy device found, and no enumeration attempt or error logged in dmesg - the kernel has no record of the device ever attempting to connect.

5. Verified the required udev rules file (/etc/udev/rules.d/49-teensy.rules) is present and correctly configured, matching the documented reference configuration exactly.

6. Verified the application-level Teensy configuration (/etc/teensyconfig) is present and well-formed valid JSON (button mappings, callbacks, title).

7. Checked the status of the teensycfg service, which applies the display/button configuration to the board: the service is in a failing auto-restart loop (exit code 1), consistent with it repeatedly failing to open the (non-existent) serial device.

Conclusion: all software and configuration layers relevant to the Teensy Board (udev rules, JSON configuration, control service) were verified correct, yet the board is completely absent from USB enumeration at the kernel level. This rules out a software or configuration cause and points to a physical/electrical fault - most likely a disconnected or damaged internal USB cable/connector, or physical damage to the Teensy Board itself - consistent with the reported impact. Physical inspection of the internal USB connection to the Teensy Board is recommended as the next step; if the connector is intact and properly seated, replacement of the Teensy Board is recommended.

Appendix: Technical Evidence (logs and command output)

Teensy Serial Device Check

```
$ ls /dev/ttyACM*
ls: cannot access '/dev/ttyACM*': No such file or directory
```

Teensy Communication Attempt (Python control library)

```
$ python3 -c 'from teensycmds.teensycmds import TeensyBoard; t=TeensyBoard(); print(t.cmd("ping"))'

Traceback (most recent call last):
File "/usr/local/lib/python3.8/dist-packages/serial/serialposix.py", line 265, in open
self.fd = os.open(self.portstr, os.O_RDWR | os.O_NOCTTY | os.O_NONBLOCK)
FileNotFoundError: [Errno 2] No such file or directory: '/dev/ttyACM0'

During handling of the above exception, another exception occurred:

Traceback (most recent call last):
File "<string>", line 1, in <module>
File "/usr/local/lib/python3.8/dist-packages/teensycmds/teensycmds.py", line 75, in cmd
ser = serial.Serial('/dev/{}'.format(self.port), self.baudrate, ...)
serial.serialutil.SerialException: [Errno 2] could not open port /dev/ttyACM0:
[Errno 2] No such file or directory: '/dev/ttyACM0'
```

Full USB Device List (lsusb)

```
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 005: ID 1199:9091 Sierra Wireless, Inc.
Bus 001 Device 004: ID 05e3:0608 Genesys Logic, Inc. Hub
Bus 001 Device 003: ID 8087:0025 Intel Corp.
Bus 001 Device 002: ID 0403:6010 Future Technology Devices International, Ltd FT2232C/D/H Dual UART/FIFO IC
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub

Note: no device with vendor ID 16c0 (PJRC/Teensy) is present. Only the cellular
modem, a USB hub, an Intel controller, and an FTDI serial converter (unrelated
to the Teensy) are listed.
```

Kernel USB Messages (dmesg, filtered)

```
[ 4.256030] usb usb1: New USB device strings: Mfr=3, Product=2, SerialNumber=1
[ 4.274497] usb usb1: SerialNumber: 0000:00:15.0
[ 4.296074] usb usb2: New USB device strings: Mfr=3, Product=2, SerialNumber=1
[ 4.314529] usb usb2: SerialNumber: 0000:00:15.0
[ 4.709054] usb 1-2: New USB device strings: Mfr=1, Product=2, SerialNumber=0
[ 5.027122] usb 1-3: New USB device strings: Mfr=0, Product=0, SerialNumber=0
[ 5.323547] usb 1-4: New USB device strings: Mfr=0, Product=1, SerialNumber=0
[ 5.773752] usb 1-4.1: New USB device strings: Mfr=1, Product=2, SerialNumber=3
[ 5.795425] usb 1-4.1: SerialNumber: UF14163003011049
[ 10.852383] usbcore: registered new interface driver usbserial_generic
[ 10.869849] ftdi_sio 1-2:1.0: FTDI USB Serial Device converter detected
[ 10.871080] usb 1-2: FTDI USB Serial Device converter now attached to ttyUSB0
[ 10.871283] usb 1-2: FTDI USB Serial Device converter now attached to ttyUSB1
[ 10.931966] usbcore: registered new interface driver qcserial
```

```
Note: no mention of the Teensy anywhere in the boot log - not even a failed
enumeration attempt. The device never attempted to connect to the USB bus.
```

udev Rules for Teensy (/etc/udev/rules.d/49-teensy.rules)

```
ATTRS{idVendor}=="16c0", ATTRS{idProduct}=="04*", ENV{ID_MM_DEVICE_IGNORE}="1", ENV{ID_MM_PORT_IGNORE}="1"
ATTRS{idVendor}=="16c0", ATTRS{idProduct}=="04[789a]*", ENV{MTP_NO_PROBE}="1"
KERNEL=="ttyACM*", ATTRS{idVendor}=="16c0", ATTRS{idProduct}=="04*", MODE:="0666", RUN:="/bin/stty -F /dev/%k
raw -echo"
KERNEL=="hidraw*", ATTRS{idVendor}=="16c0", ATTRS{idProduct}=="04*", MODE:="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="16c0", ATTRS{idProduct}=="04*", MODE:="0666"
KERNEL=="hidraw*", ATTRS{idVendor}=="1fc9", ATTRS{idProduct}=="013*", MODE:="0666"
SUBSYSTEMS=="usb", ATTRS{idVendor}=="1fc9", ATTRS{idProduct}=="013*", MODE:="0666"

Note: rules file matches the documented reference configuration correctly -
ruled out as the cause.
```

teensyconfig Service Status

```
$ service teensycfg status

teensycfg.service - Teensy configuration service
Loaded: loaded (/etc/systemd/system/teensycfg.service; enabled; vendor preset)
Active: activating (auto-restart) (Result: exit-code)
Process: 17451 ExecStart=/usr/local/bin/teensycfg --load --apply (code=exited, status=1/FAILURE)
Main PID: 17451 (code=exited, status=1/FAILURE)

Note: service is stuck in a failing restart loop, consistent with repeated
failed attempts to open the non-existent /dev/ttyACM0 device.
```

Teensy Application Configuration (/etc/teensyconfig)


```
{
"buttons": {
"1": ["inop"],
"2": ["inop"],
"3": ["inop"],
"4": ["inop"],
"5": ["wifi_off", "wifi_on"]
},
"callbacks": {
"wifi_off": "led5off",
"wifi_on": "led5on"
},
"wap": "on",
"title": "Avianca Airlines"
}
```


```
Note: valid, well-formed configuration - ruled out as the cause.
```